

PRESENTATION

Lung Cancer Prediction by Using Data Mining Techniques

Data Mining for Business

Group Members:

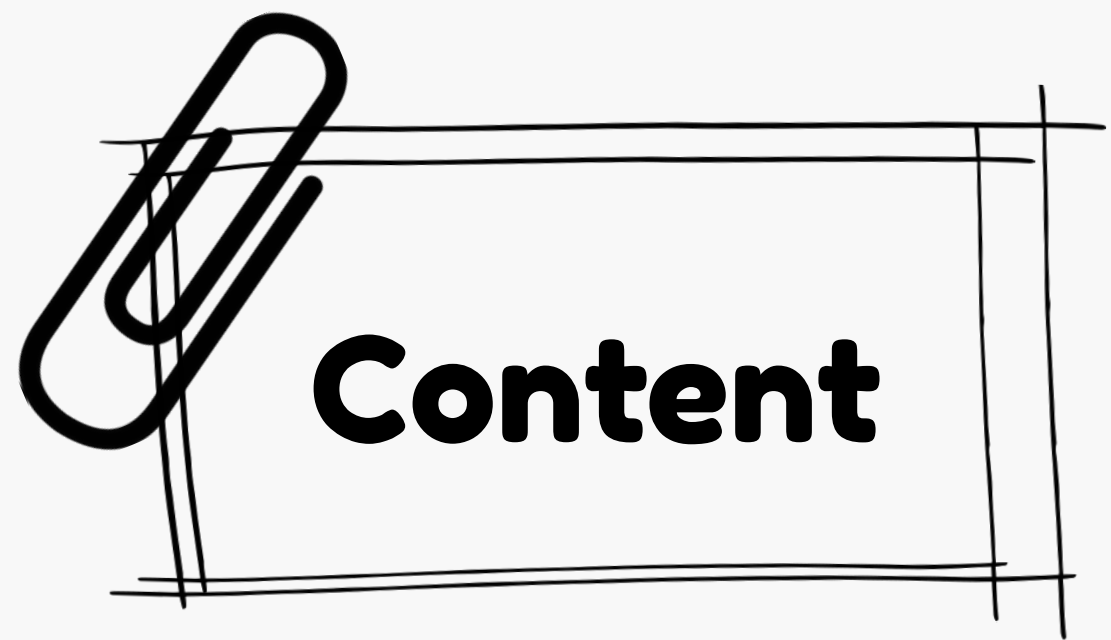
Wenting XU

Chen HE

Tingyi KAO

Hein Htet SOE THAN





Content

Dataset description

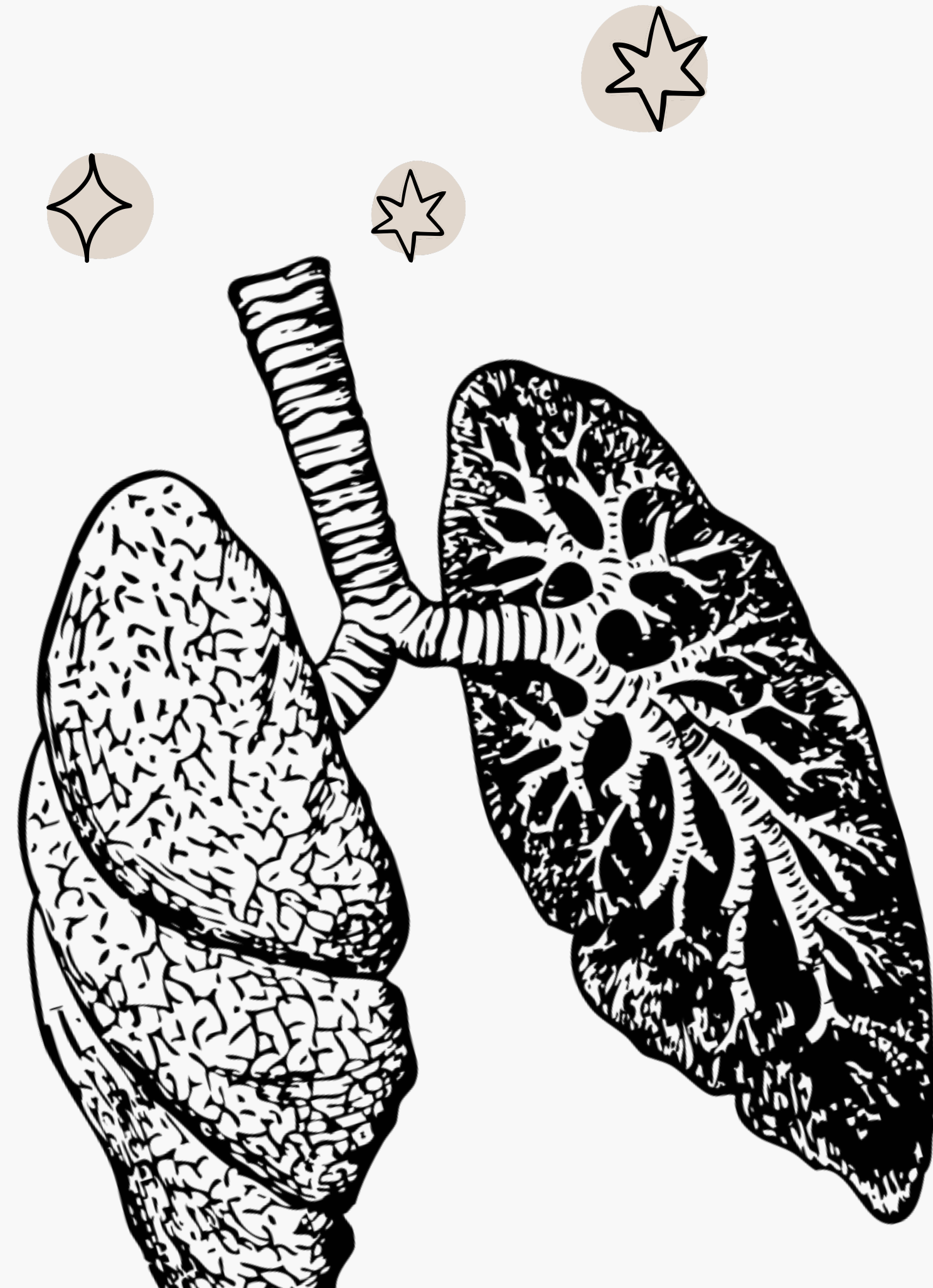
Objective of the study

Theoretical background

Results and interpretation

Conclusions

1



Dataset description

Dataset Overview

1,000

Patients

N = sample size

24

Predictor Variables

4 thematic categories

3

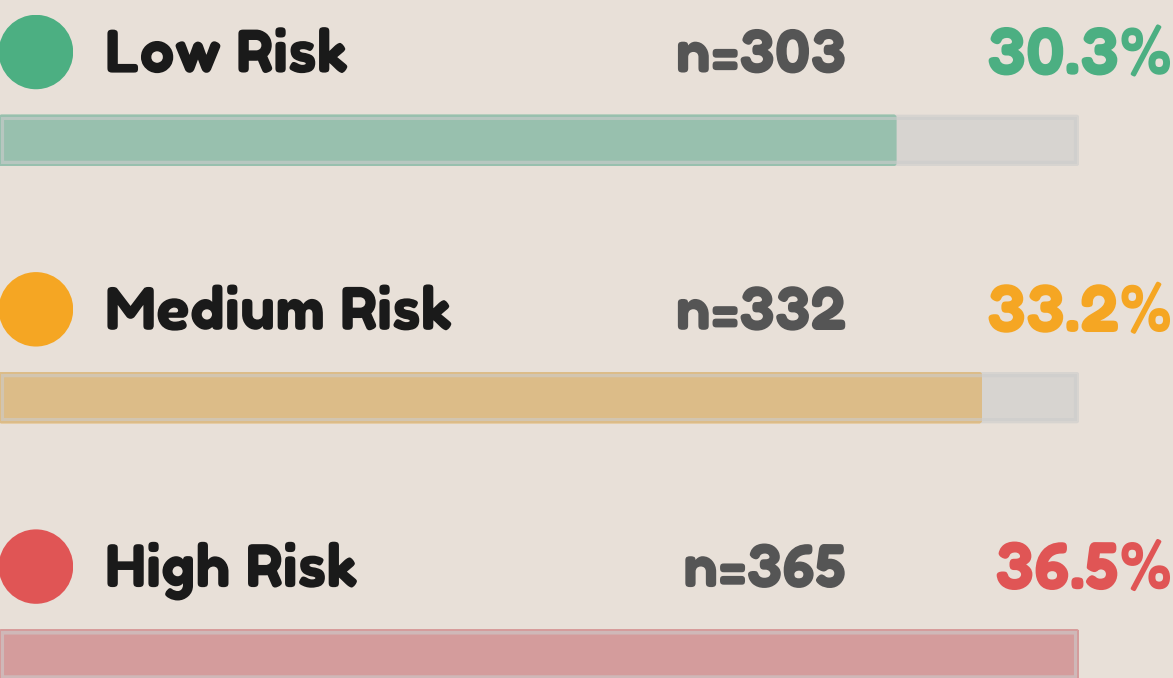
Risk Levels

Low · Medium · High

Data Source & Structure

Dataset Title:	Cancer Patient Risk Assessment
Unit of Analysis:	Individual patient record (one row = one patient)
Data Structure:	Cross-sectional observational data
Scale Types:	Ordinal (1–8), (1–7), (1–9) for risk factors & symptoms
Missing Values:	None – complete data for all 1,000 patients

Outcome Distribution



Variable Categories

24 predictor variables + 1 outcome variable

Demographics

02

N = 2 variables

- Age – 14 to 73 years (Mean: 37.2)
- Gender – Male 59.8% | Female 40.2%

Environmental & Lifestyle

06

Scale: 1–8 ordinal

- Air Pollution
- Alcohol Use
- Dust Allergy
- Occupational Hazards
- Smoking
- Passive Smoker

Medical History & Health

04

Scale: 1–7 ordinal

- Genetic Risk
- Chronic Lung Disease
- Balanced Diet
- Obesity

Clinical Symptoms

11

Scale: 1–9 ordinal

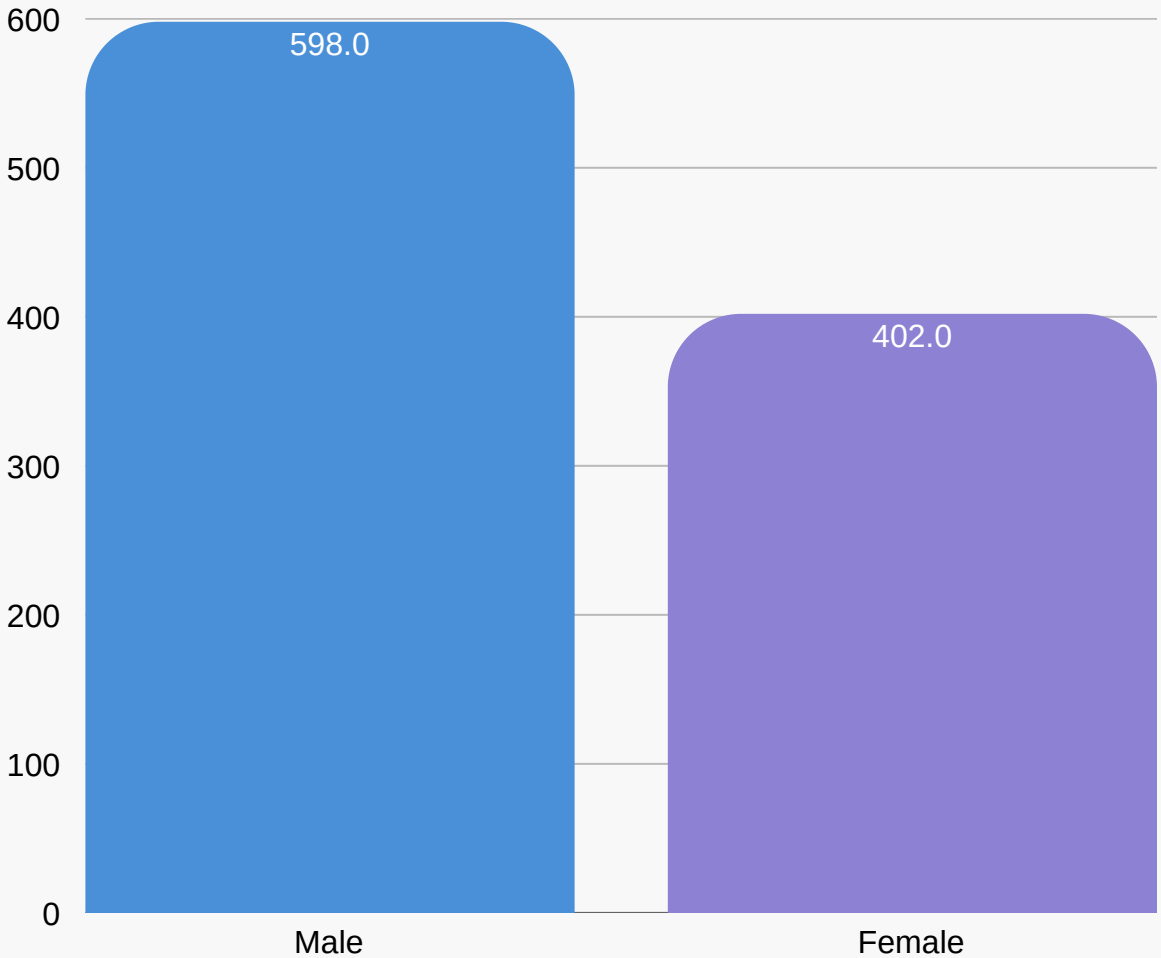
- | | |
|-----------------------|----------------------------|
| • Chest Pain | • Clubbing of Finger Nails |
| • Coughing of Blood | • Dry Cough |
| • Fatigue | • Swallowing Difficulty |
| • Weight Loss | • Frequent Cold |
| • Shortness of Breath | • Snoring |
| • Wheezing | |

Demographics & Risk Distribution

Age Distribution

Mean	37.2 yrs
Median	36.0 yrs
Std Dev	12.0 yrs
Range	14 – 73 yrs
IQR	28 – 45 yrs

Gender Distribution



Cancer Risk Level Distribution

Low (30.3%) Medium (33.2%) High (36.5%)



Age is approximately symmetric (mean $\approx$ median), spanning adolescence to elderly adults.

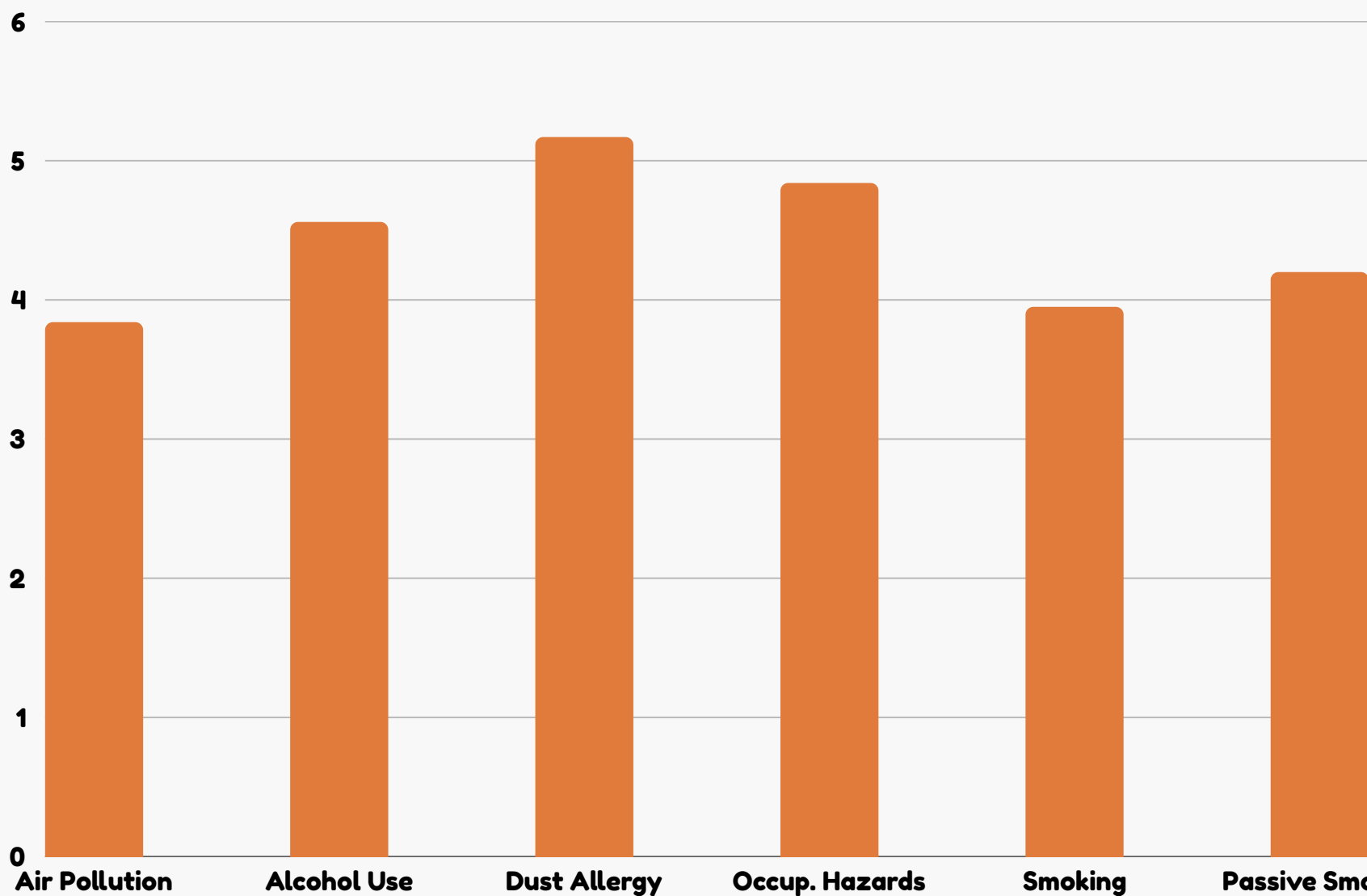
60% male — gender imbalance should be considered.

Balanced outcome classes (min 30.3%) are ideal for classification models.

Descriptive Statistics

Environmental & Lifestyle Factors · Medical History & Health Conditions

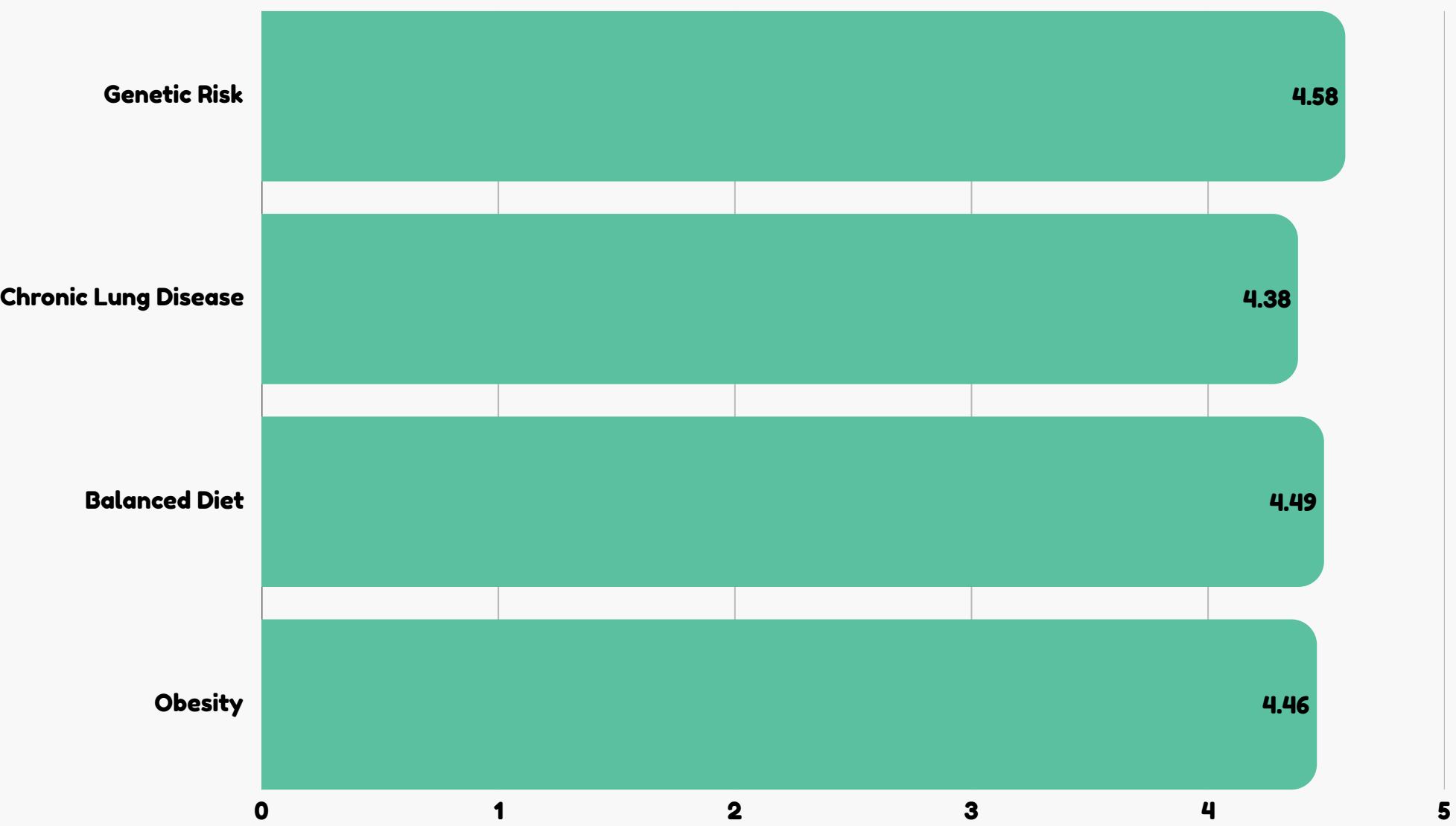
Environmental & Lifestyle Risk Factors (Mean Severity)



Dust Allergy Highest

Mean 5.17/8 – highest exposure among lifestyle factors, indicating significant airborne risk.

Medical History & Health Conditions (Mean Score)



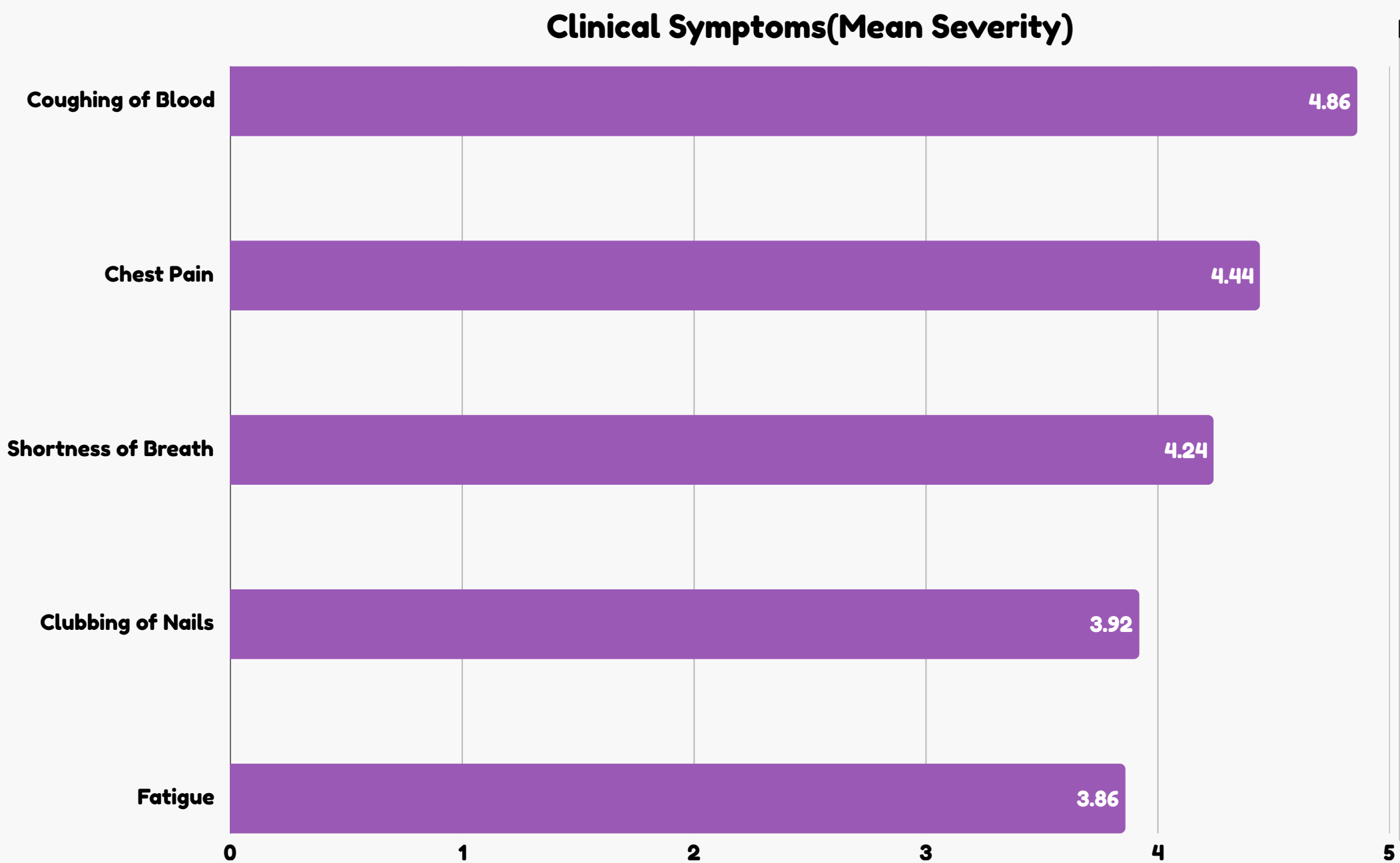
Genetic Risk Dominant

Mean 4.58/7 – substantial hereditary predisposition across the sample population.

Clustered Medical Scores

All four medical factors cluster tightly (4.38–4.58), suggesting uniformly moderate pre-existing health risk.

Clinical Symptoms & Exploratory Preview



Exploratory Analysis Preview

MCA

Multiple Correspondence Analysis to explore relationships among categorical risk variables and patient profiles

CLU

Clustering Analysis to detect natural patient subgroups based on symptom profiles

Coughing of Blood (4.86)

Most severe symptom – critical clinical indicator

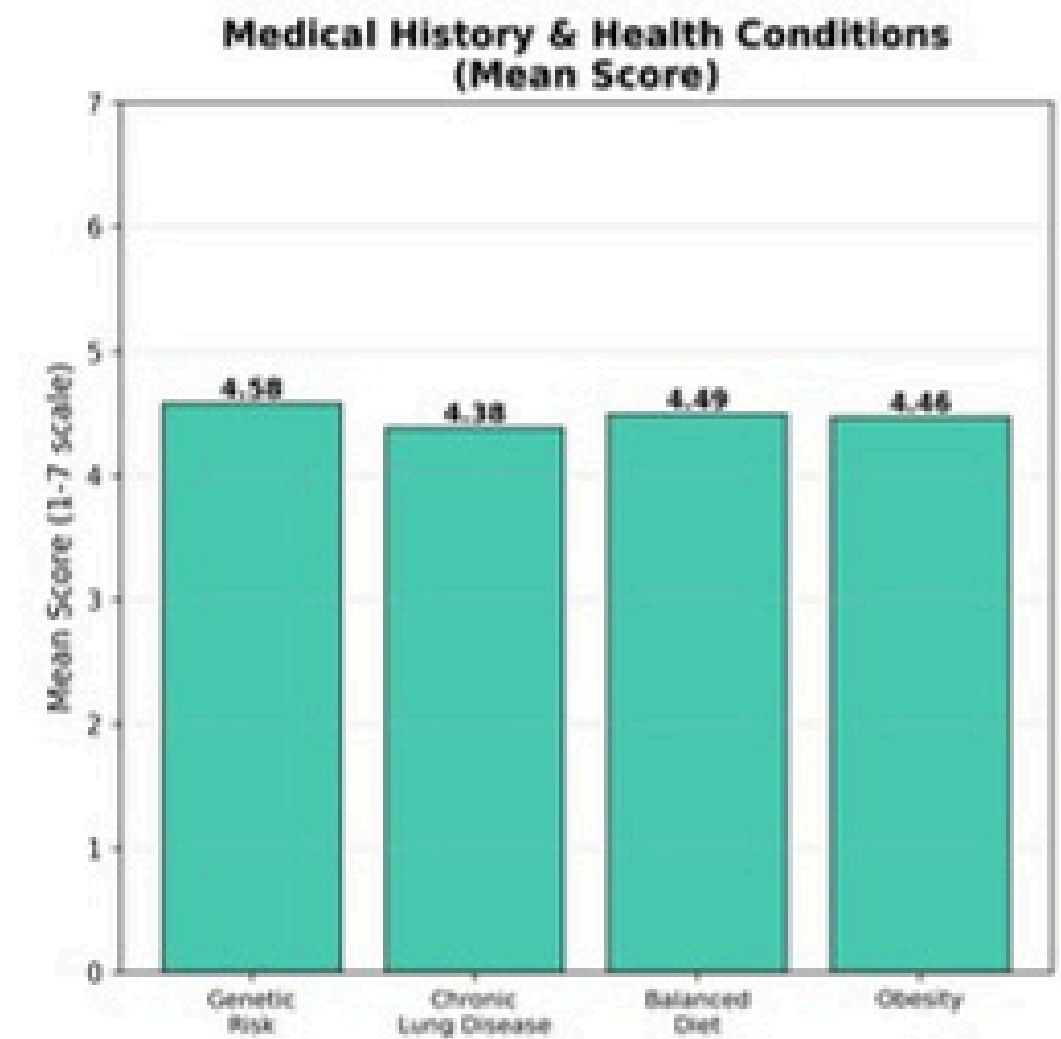
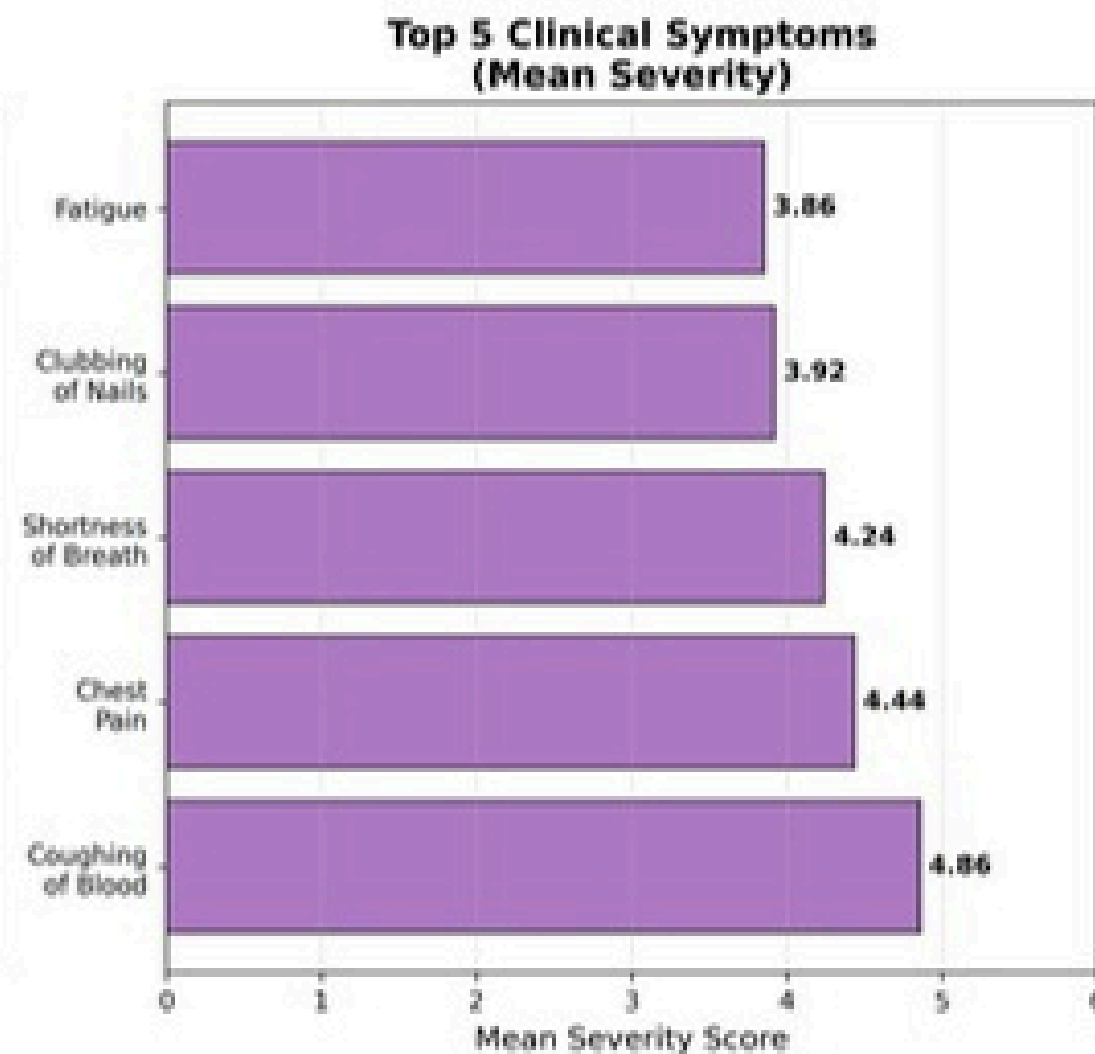
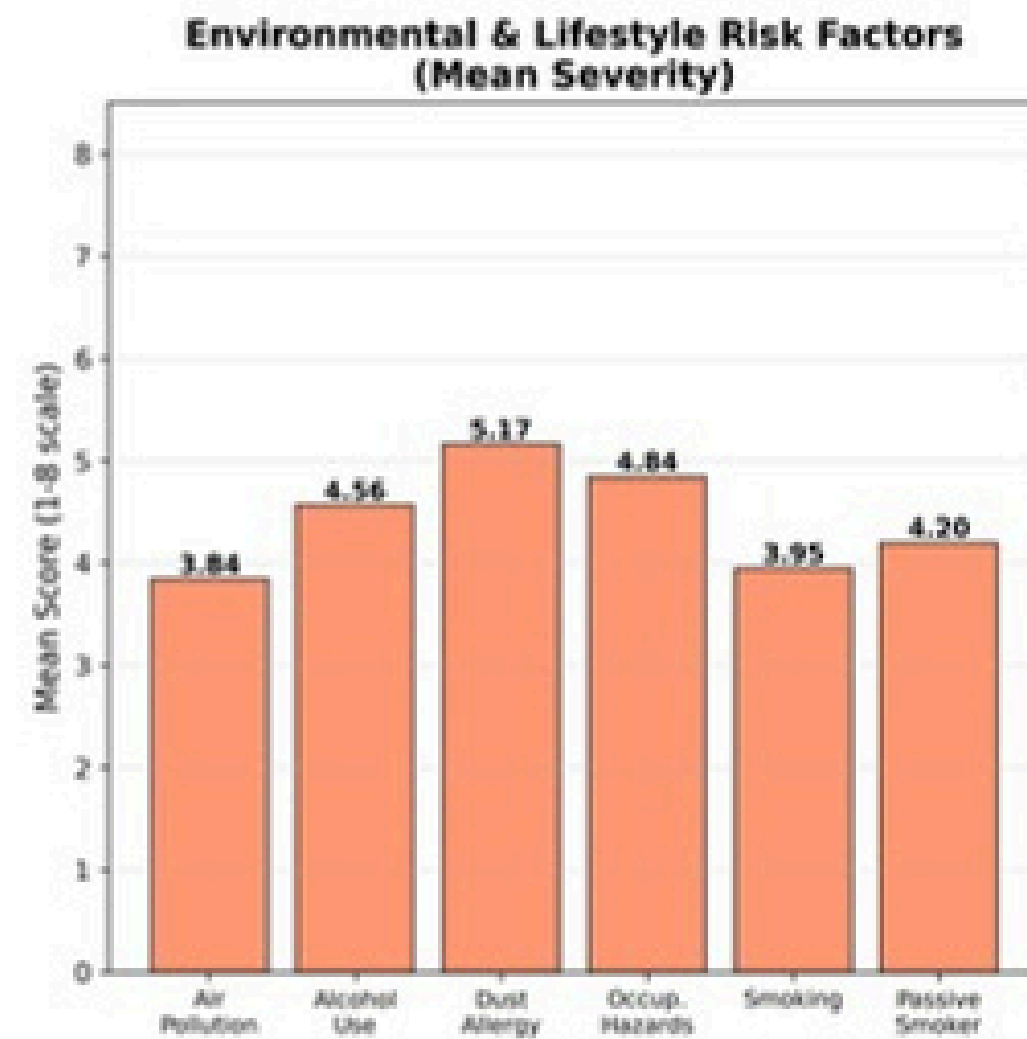
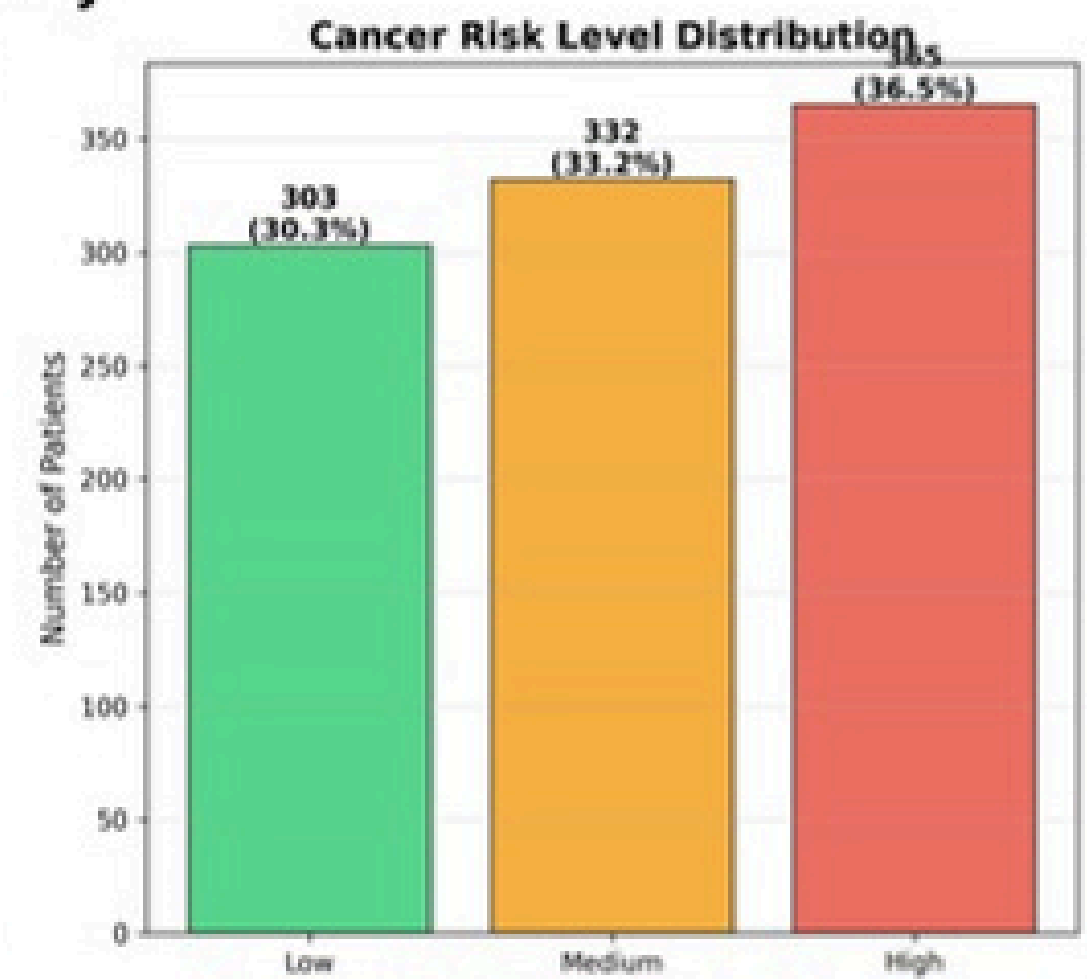
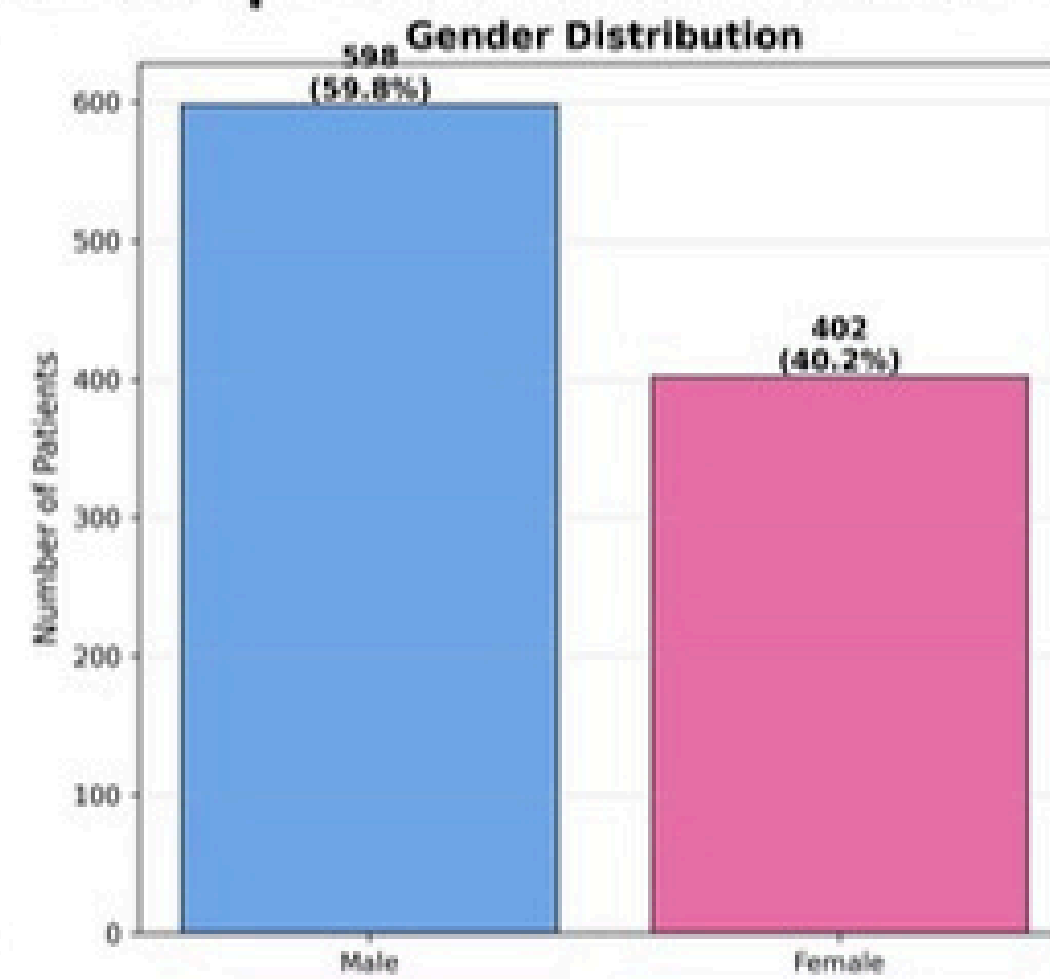
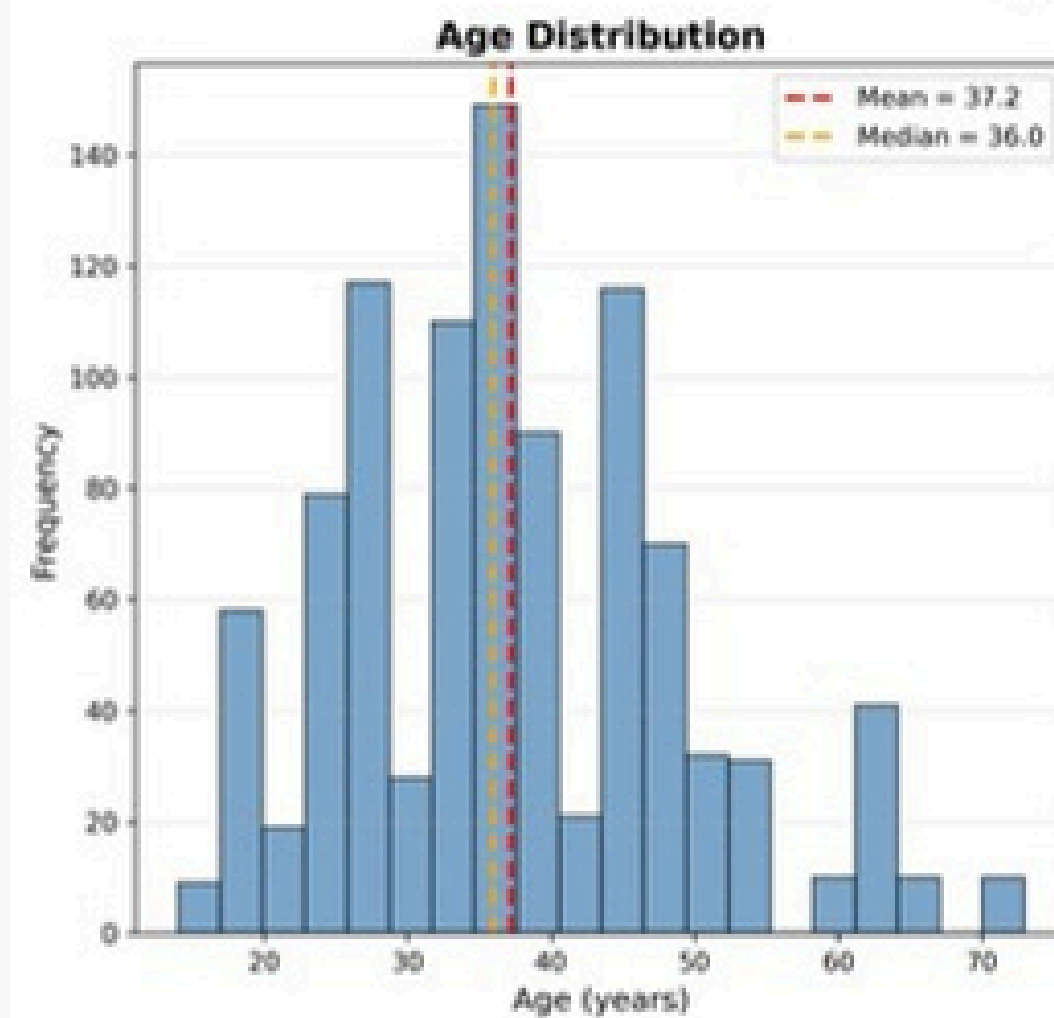
Respiratory Focus

4 of top 5 symptoms are cardiopulmonary(heart and lung related symptoms)

High Variability

SD 2.0–2.4 across symptoms – diverse patient profiles

Basic Descriptive Statistics - Visual Summary



Objective of the study

Research Objectives

Objective

To identify patient profiles associated with lung cancer risk factors using **Multiple Correspondence Analysis** and **Clustering** techniques.

Are certain risk factors (such as smoking, air pollution) associated with specific lung cancer symptoms?

Question 1 

Question 2 

Can we identify groups of patients with similar profiles based on their exposure to risk factors and reported symptoms?

What patterns exist in the dataset that may help better understand the interaction between lifestyle, environment, and lung cancer indicators?

Question 3 

Research Method Justifications

MCA

1. **Reduce the high dimensionality** of the dataset
2. Transform categorical variables into **a smaller number of synthetic dimensions**
3. **Visualize the relationships** between categories and individuals
4. **Identify associations** between risk factors and symptoms

Hierarchical Clustering

1. Does not require specifying the number of clusters in advance
2. Provides a **dendrogram**, which visually represents the structure of the clusters
3. Helps **identify natural groupings** of patients

By combining MCA with hierarchical clustering, we might know

1. Reduce the complexity of the categorical dataset
2. Identify clusters of patients with similar profiles
3. Interpret these clusters in terms of risk factors and symptoms associated with lung cancer

Theoretical background

**This section briefly introduces the two
core statistical methods we used:
Multiple Correspondence Analysis
(MCA) and Hierarchical Clustering (HC)**

Multiple Correspondence Analysis (MCA)

A dimensionality reduction tool specifically for categorical data (e.g, symptom levels, gender) that visualizes complex relationships on a simple plot.

Core Purpose:

When there are too many variables (e.g, 20+ symptoms) to analyze easily, MCA compresses them into 2-3 new dimensions while keeping as much of the original information as possible.

Key Concepts:

Inertia: Like the "spread" of the data. We want the first two dimensions to "explain" as much of this spread as possible.

Contribution:

Tells us which symptom (e.g, "Coughing") is most important for defining a dimension.

Special Step in Our Code:

We set "Cancer Level (Level)" as a supplementary variable (a "bystander").

What this means:

It did not influence how the axes were drawn. It was only added to the plot at the end.

Why we did it:

To test if our model could automatically separate "mild" and "severe" patients without being told the cancer level beforehand.

Hierarchical Clustering(HC)

An algorithm that automatically groups "similar patients" together without needing to be told what the groups are.

Core Purpose:

After MCA simplified the data, we wanted to see how many distinct groups of patients there were.

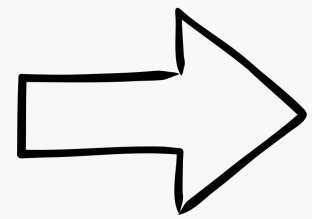
Our Step-by-Step Process:		
Step 1	Input Data	We used the coordinates from the first 5 MCA dimensions (this is cleaner and removes noise)
Step 2	Calculate Distance	We used "Euclidean distance" to measure how similar or different each patient was from the others
Step 3	Clustering Method	Ward.D2 method: This is the most common method because it creates groups where the members are as similar as possible.
Step 4	Draw a Dendrogram	The result looks like a family tree, showing how patients merge into groups step by step.
Step 5	Pick the Number of Groups	We used two methods to decide between 3 or 4 clusters: Elbow Method & Silhouette Method



Results and interpretation

Multiple Correspondence Analysis (MCA)

In MCA, categorical variables must be transformed into binary variables.



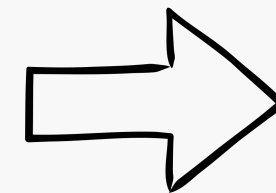
Complete Disjunctive Coding

24 variables (P) $\longrightarrow$ 174 columns (S)
binary variables

Gender		Air.Pollution							
1	2	1	2	3	4	5	6	7	8
1	0	0	1	0	0	0	0	0	0
1	0	0	0	1	0	0	0	0	0
1	0	0	0	0	1	0	0	0	0

The Burt table is obtained by multiplying the disjunctive matrix by its transpose.

	1	2	1	2	3	4	5	6	7	8
1	598	0	42	128	90	38	10	268	12	10
2	0	402	0	73	83	52	10	58	18	9
1	42	99	141	0	0	0	0	0	0	0
2	128	73	0	201	0	0	0	0	0	0
3	90	83	0	0	173	0	0	0	0	0
4	38	52	0	0	0	90	0	0	0	0
5	10	10	0	0	0	0	20	0	0	0
6	268	58	0	0	0	0	0	326	0	0
7	12	18	0	0	0	0	0	0	30	0
8	10	9	0	0	0	0	0	0	0	19



It summarizes all contingency tables between categories of the variables.

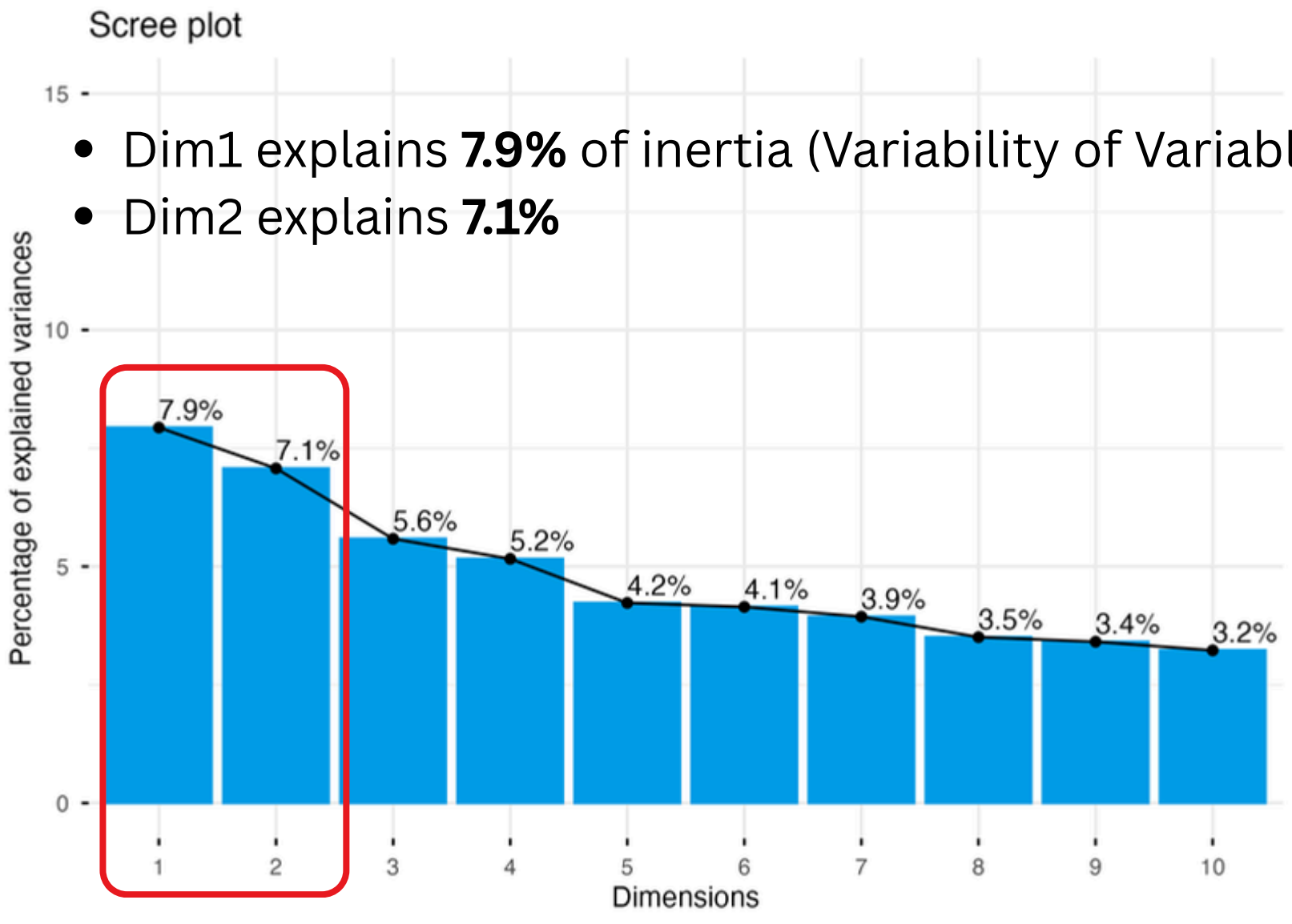
- 598 individuals are Gender 1
- 402 individuals are Gender 2
- 42 individuals are Gender 1 and Air Pollution level 1
- 128 individuals are Gender 1 and Air Pollution level 2
- 99 individuals are Gender 2 and Air Pollution level 1

“Male”
“Female”

= 1000 (total observations)

Multiple Correspondence Analysis (MCA)

Eigenvalues and Explained Inertia

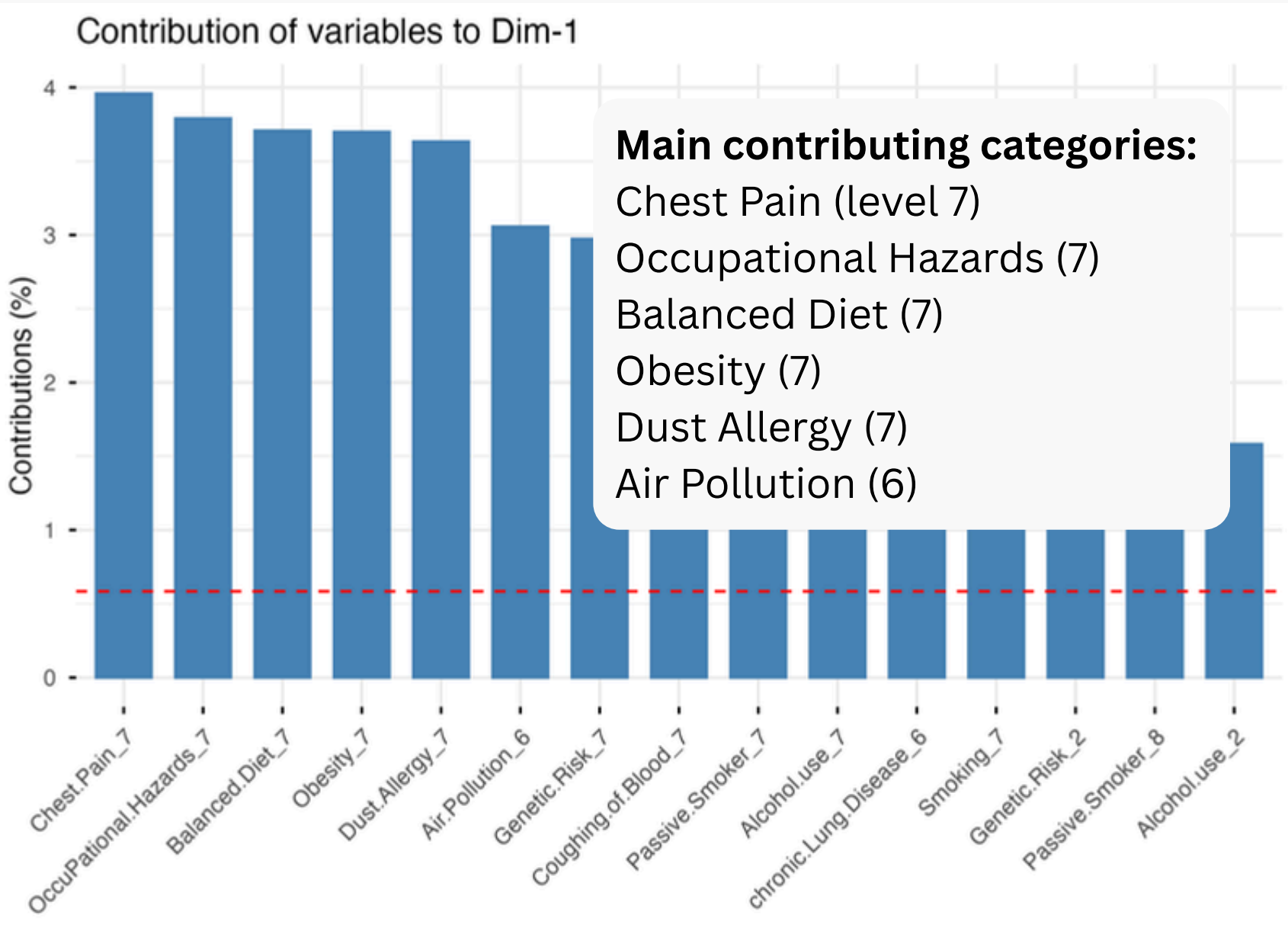


	eigenvalue	variance.percent	cumulative.variance.percent
Dim.1	0.51	7.94	7.94
Dim.2	0.46	7.07	15.01
Dim.3	0.36	5.59	20.60
Dim.4	0.33	5.16	25.76
Dim.5	0.27	4.23	29.99
Dim.6	0.27	4.15	34.14
Dim.7	0.25	3.94	38.08
Dim.8	0.23	3.51	41.58
Dim.9	0.22	3.41	44.99
Dim.10	0.21	3.23	48.22

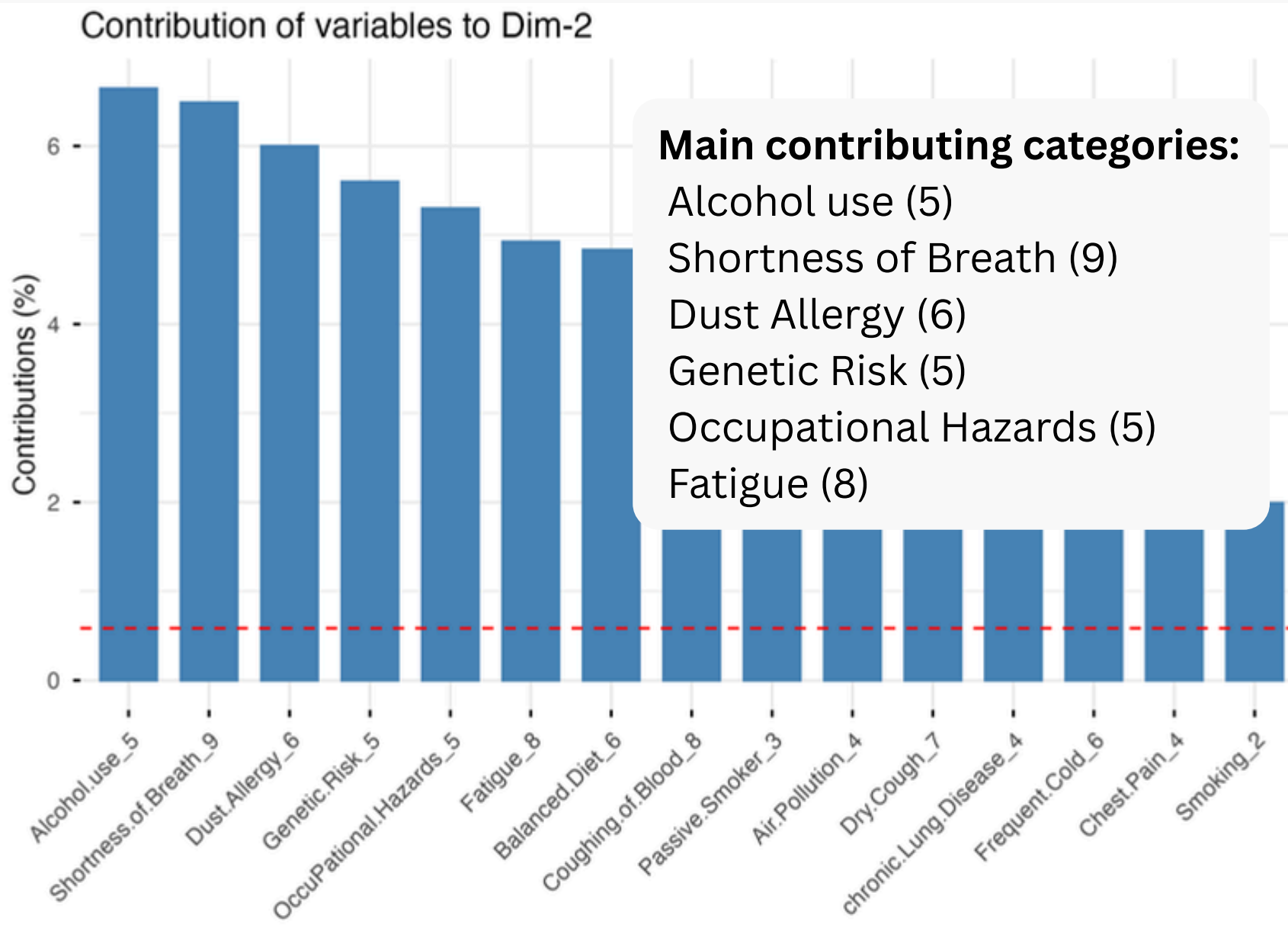
- ➔
- First two dimensions capture the main structure of the data.
 - The scree plot shows a gradual decline, indicating diminishing information gain after the first dimensions.

Multiple Correspondence Analysis (MCA)

Risk Exposure & Health Conditions

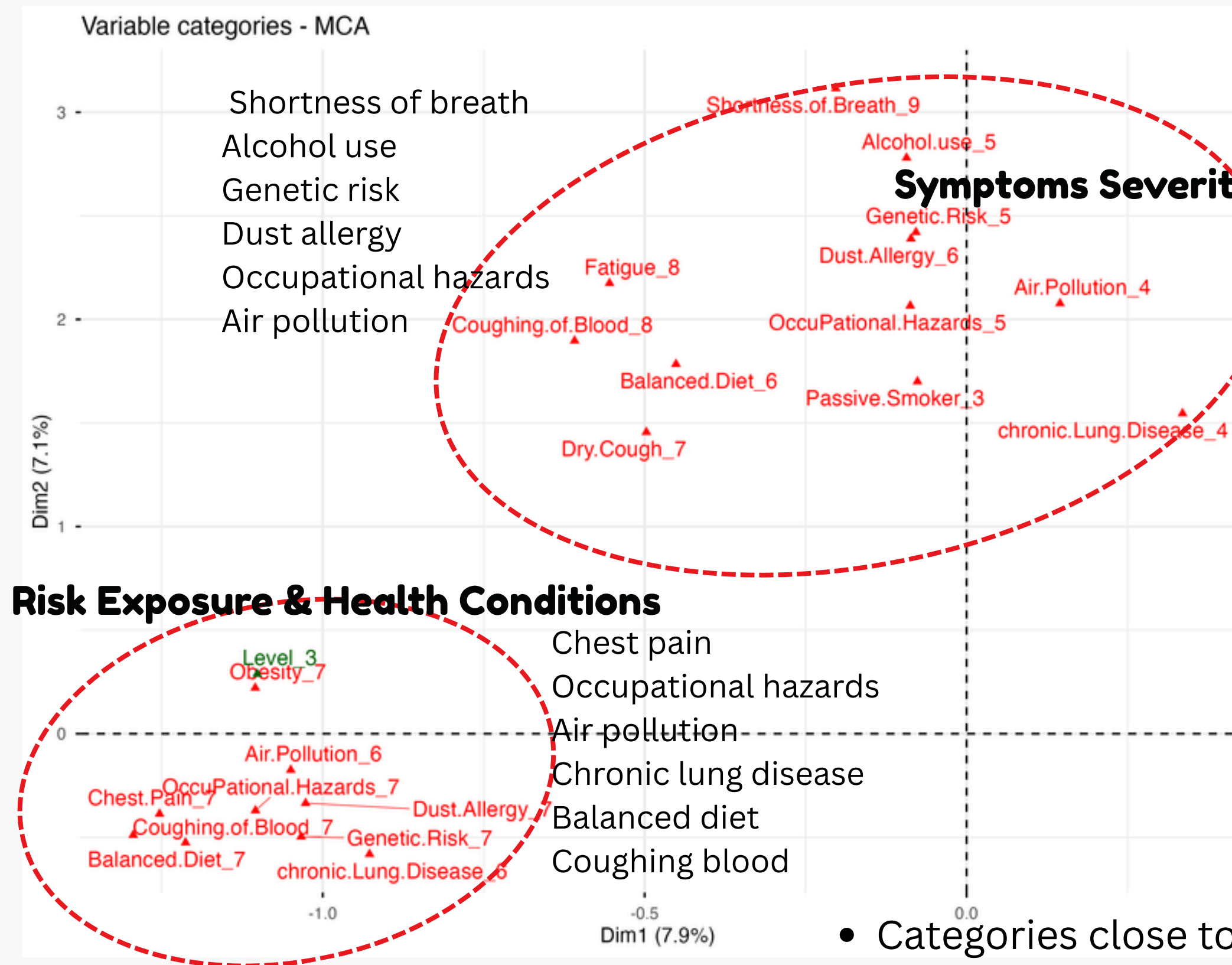


Symptoms Severity



- Dim1 reflects exposure to environmental and health risk factors (air pollution, occupational hazards, obesity).
- Dim2 represents severity of respiratory symptoms (shortness of breath, fatigue).

Multiple Correspondence Analysis (MCA)

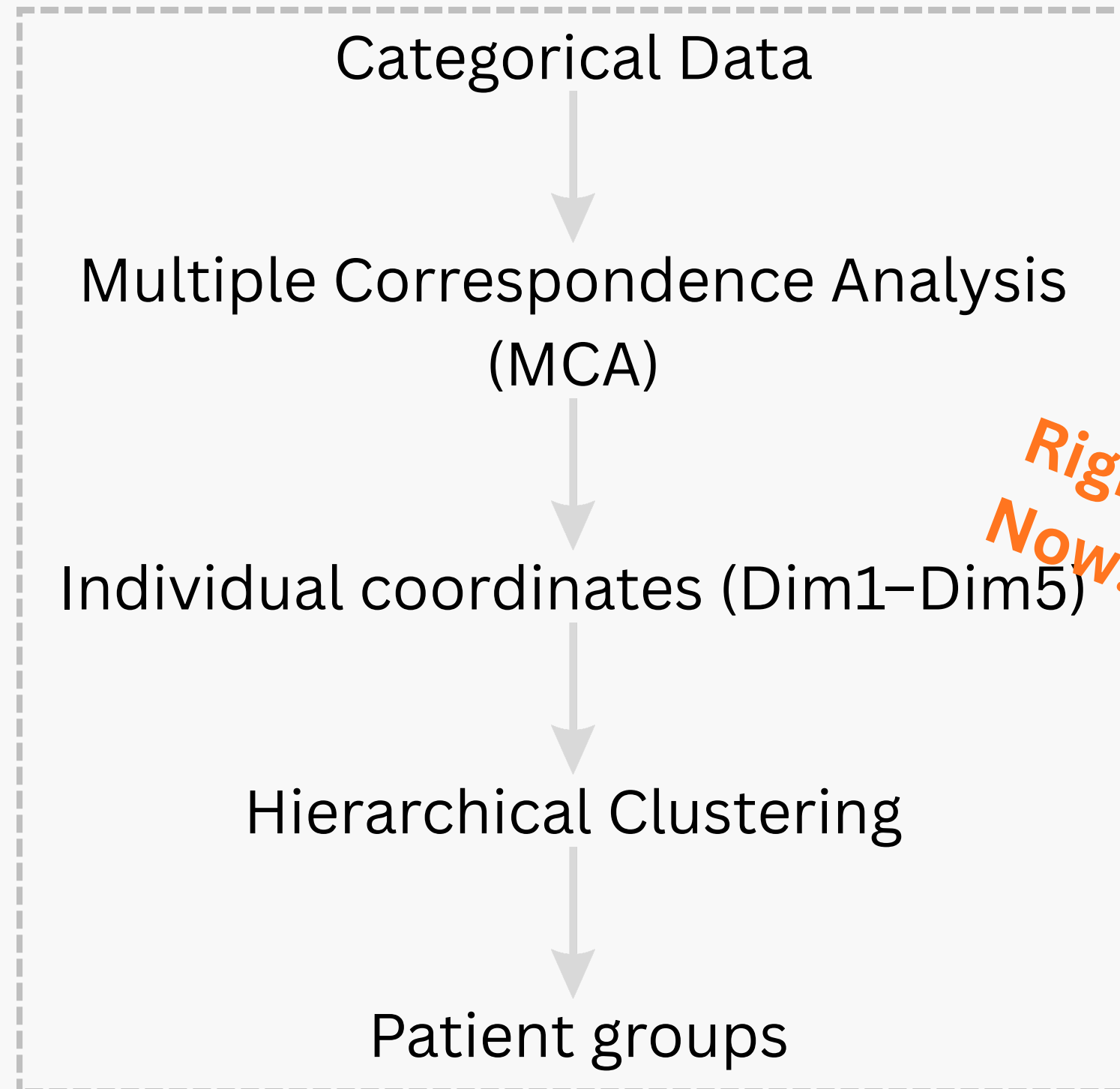


- The MCA variable map includes only categories with $\cos^2 \geq 0.4$ to improve interpretability.

- Categories close together → associated
- Categories far from origin → stronger contributors to the dimension
- Opposite sides → contrasting patterns

Hierarchical Clustering

What We Have Done So Far



Right Now!!

The coordinates of individuals on the MCA dimensions are used as input for hierarchical clustering.

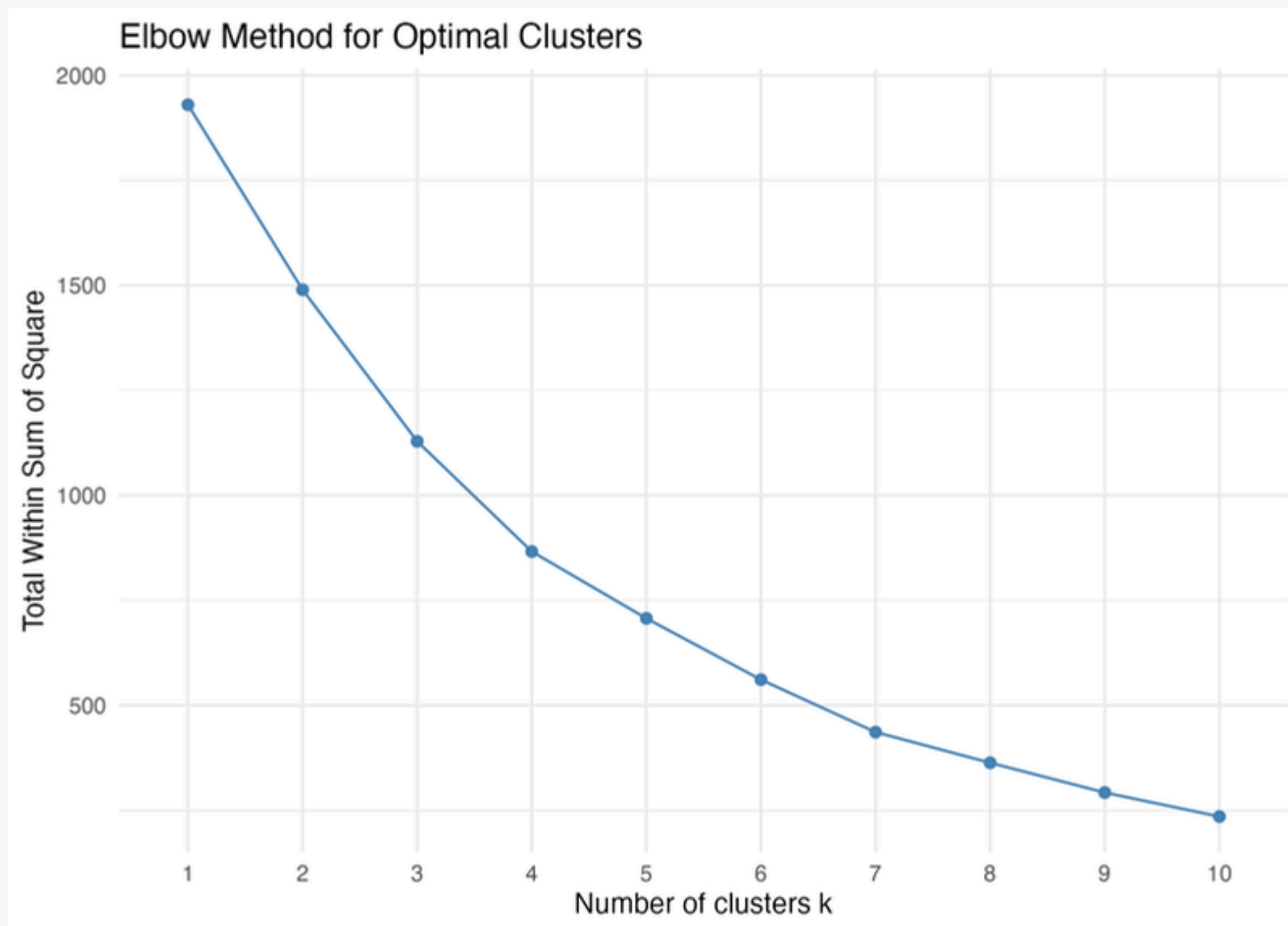
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The first five MCA dimensions were retained as input for clustering, as they capture the main variability of the data (around 30% of total inertia).

Hierarchical Clustering

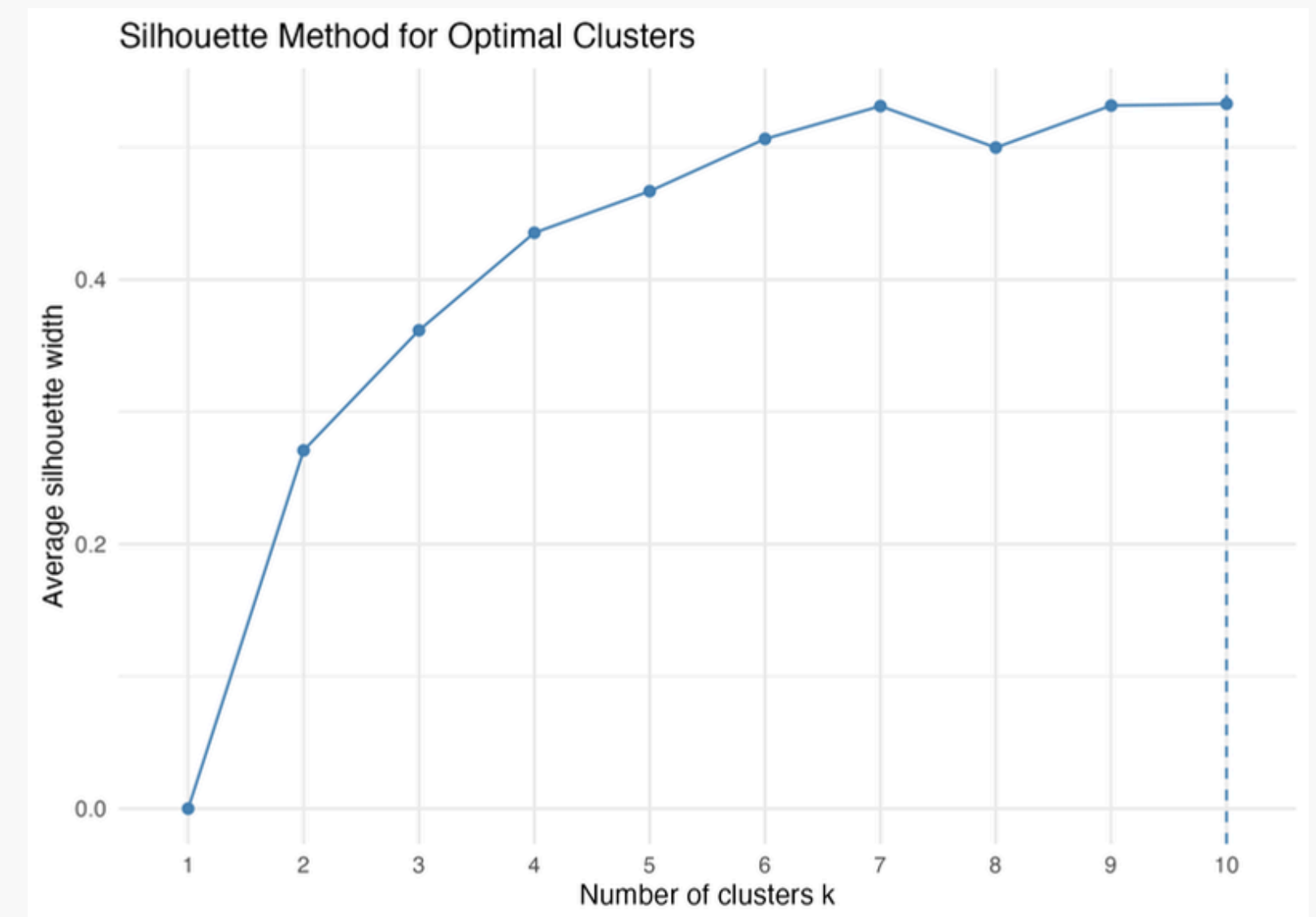
Elbow Method

WSS decreases rapidly until $k \approx 3-4$
After that, improvement becomes marginal



Silhouette Method

Measures cluster separation and cohesion
Higher value means better clustering
Scores improve up to $k \approx 3-4$



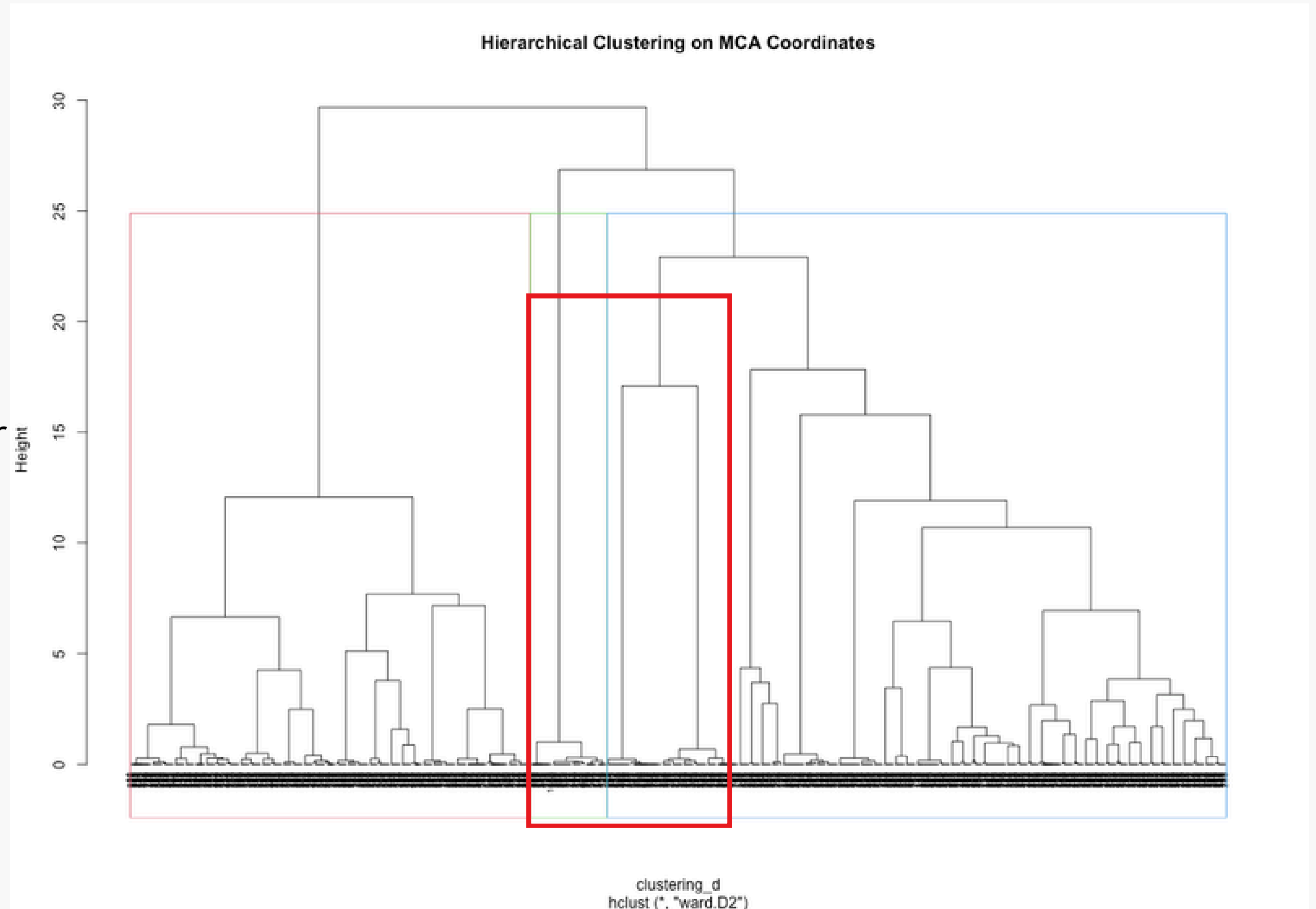
Hierarchical Clustering

- Input: first 5 MCA dimensions
- Distance: Euclidean
- Linkage: Ward.D2

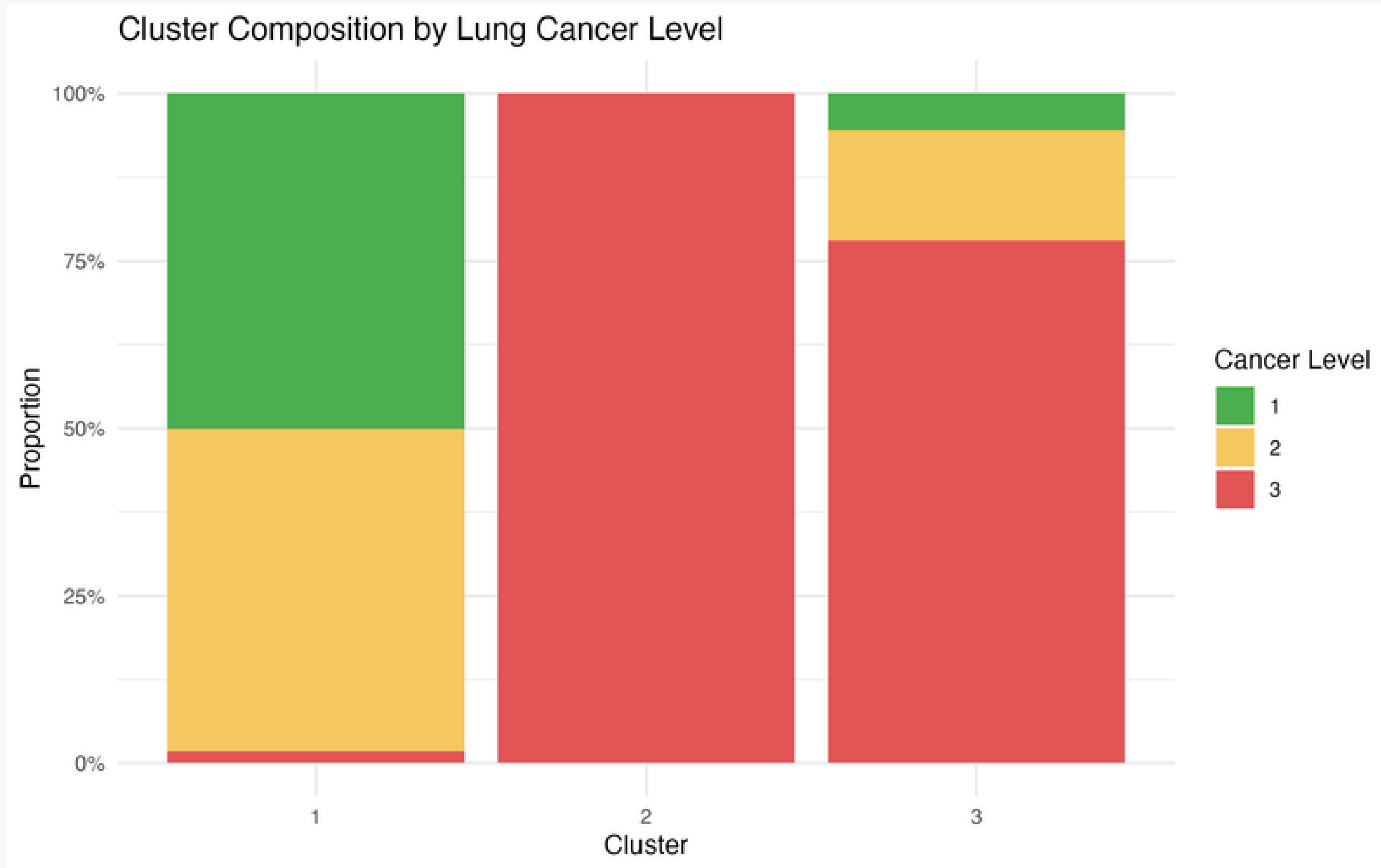
Why We Choose k=3

1. Large jumps in fusion height suggest a small number of groups
2. A 3-cluster solution provides a clear and interpretable partition
3. It balances simplicity and heterogeneity

Clusters	Frequency
1	565
2	70
3	365



Clustering Interpretation



Cluster 1 – Moderate Risk Profile

- Roughly half Level 1 and half Level 2
- Very few Level 3 cases
- Represents patients with moderate symptoms or exposure

Cluster 2 – Extreme High Risk Profile

- Almost entirely Level 3
- Indicates patients with the most severe lung cancer indicators
- Strong association between this cluster and high risk

Cluster 3 – High Risk Profile

- Majority Level 3, with some Level 2
- Small proportion of Level 1
- Represents patients with elevated risk factors

Clustering Interpretation

Air Pollution

Clusters	1	2	3	4	5	6	7	8
1	0.143	0.32	0.306	0.071	0.035	0.124	0	0
2	0	0	0	0.714	0	0.286	0	0
3	0.164	0.055	0	0	0	0.647	0.082	0.052

Smoking

Clusters	1	2	3	4	5	6	7	8
1	0.32	0.269	0.198	0.035	0.018	0.088	0.053	0.018
2	0	1	0	0	0	0	0	0
3	0	0	0.164	0.107	0	0.027	0.485	0.216

Occupational Hazards

Clusters	1	2	3	4	5	6	7	8
1	0.088	0.234	0.25	0.198	0.053	0.053	0.124	0
2	0	0	0	0	1	0	0	0
3	0	0	0.027	0	0.082	0	0.808	0.082

Conclusions

Summary of Key Findings (MCA)

Dimensionality Reduction

The first two dimensions (Dim1 & Dim2) capture approximately 15% of the total inertia



Dimension 1: Risk Exposure & Health Conditions

This dimension is most strongly influenced by lifestyle and environmental factors.

- High Severity Risk Factors: Obesity (Level 7), Occupational Hazards (Level 7), and Poor Balanced Diet (Level 7).
- Environmental Impact: Air Pollution (Level 6) also plays a significant role in defining this axis.

Dimension 2: Symptoms Severity

This dimension focuses on the active clinical state of the patient.

- Critical Symptoms: Shortness of Breath (Level 9) and Fatigue (Level 8) are the strongest contributors.
- Medical History: Genetic Risk (Level 5) and Alcohol Use (Level 5) also align with this axis.

Summary of Key Findings (Clustering)

Cluster	Risk Profile	Key Findings & Indicators	Actionable Strategies
Cluster 1 (565)	Moderate Risk	Balanced distribution between Low and Medium risk. Minimal Level 3 cases.	Preventive Monitoring: Implement regular health check-ups and lifestyle counseling to prevent progression to high-risk levels.
Cluster 2 (70)	High Risk	Almost 100% High Risk. Strongest association with specific indicators , notably Smoking and Occupational Hazards.	Urgent Intervention: Immediate clinical screening and aggressive smoking cessation programs. Prioritize these individuals for advanced diagnostic testing (e.g., CT scans).
Cluster 3 (365)	Mixed Profile	Majority High Risk with some Medium Risk. Represents a complex interaction of elevated risk factors.	Targeted Assessment: Conduct comprehensive symptom mapping focusing on respiratory severity (e.g., coughing of blood, shortness of breath).

Discussion & Insights



1.Targeted Early Screening Strategy

- Prioritize Cluster 2 Profiles: Individuals matching the Cluster 2 profile should undergo high-frequency clinical screening immediately, even if currently asymptomatic.
- Statistical Alignment: Data analysis shows that this group's characteristics align almost 100% with High-Risk (Level 3) cancer indicators.



2.Environmental & Workplace Intervention

- Critical Risk Factors: "Dust Allergy" and "Occupational Hazards" show high mean severity scores across the dataset.
- Actionable Policy: Industries associated with these risks must enforce stricter labor protection measures and enhanced respiratory safety protocols.



Discussion & Insights



3. Gender-Specific Risk Assessment

- Demographic Imbalance: The study sample consists of approximately 60% male participants.
- Analytical Consideration: Risk exposure analysis must account for this gender distribution to avoid potential bias in generalized health recommendations.



4. Symptom-Led Monitoring

- High-Impact Indicators: "Coughing of Blood" (Mean: 4.86) and "Chest Pain" (Mean: 4.44) are the most severe clinical indicators identified.
- Diagnostic Focus: These respiratory symptoms should be treated as primary triggers for advanced diagnostic pathways.

**Secrets to living
to 100**



THANK YOU

